

Revisiting Source-Free Domain Adaptation: Insights into Representativeness, Generalization, and Variety

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Abstract

Domain adaptation addresses the challenge where the distribution of target inference data differs from that of the source training data. Recently, data privacy has become a significant constraint, limiting access to the source domain. To mitigate this issue, Source-Free Domain Adaptation (SFDA) methods bypass source domain data by generating source-like data or pseudo-labeling the unlabeled target domain. However, these approaches often lack theoretical grounding. In this work, we provide a theoretical analysis of the SFDA problem, focusing on the general empirical risk of the unlabeled target domain. Our analysis offers a comprehensive understanding of how representativeness, generalization, and variety contribute to controlling the upper bound of target domain empirical risk in SFDA settings. We further explore how to balance this trade-off from three perspectives: sample selection, semantic domain alignment, and a progressive learning framework. These insights inform the design of novel algorithms. Experimental results demonstrate that our proposed method achieves state-of-the-art performance on three benchmark datasets—Office-Home, DomainNet, and VisDA-C—yielding relative improvements of 3.2%, 9.1%, and 7.5%, respectively, over the representative SFDA method, SHOT.

1. Introduction

Domain adaptation (DA) [24, 50] has seen widespread success in various machine learning and computer vision tasks, including image recognition [44, 49], segmentation [3], object detection [14], and facial recognition [46, 47]. Several prominent approaches, such as adversarial learning [8, 48], self-supervised learning [32, 43, 45], and self-training [26, 51], have demonstrated significant improvements in these domains. However, with the growing empha-

sis on data privacy, particularly in response to regulations like the General Data Protection Regulation (GDPR) in Europe, there is an increasing challenge in ensuring individuals' data is protected. This shift toward privacy-conscious practices poses a unique obstacle for traditional DA methods, as they often rely on direct access to source domain data. In this context, many conventional DA techniques lose their efficacy, making it imperative to develop new strategies that bypass the need for source domain data while still maintaining competitive performance in target domain tasks.

Source-Free Domain Adaptation (SFDA)[40] addresses the challenge of domain adaptation under strict privacy constraints, utilizing only a pre-trained model on the source domain and unlabeled data from the target domain. To overcome these limitations, various SFDA methods have been proposed, focusing primarily on data generation[4, 12], self-training [17, 39], and self-supervised learning [2, 42]. Data generation-based methods face the challenge of effectively leveraging generated data to accurately capture the characteristics of the unseen source domain. On the other hand, self-training methods often suffer from the issue of noisy pseudo-labels, in addition to the increased computational overhead and the requirement for carefully defined pretext tasks. Despite the success of these approaches, they are often driven by intuitive motivations rather than grounded in theoretical analysis, leaving the exploration of SFDA's fundamental limits underexplored and unstructured.

To gain a deeper understanding of the Source-Free Domain Adaptation (SFDA) problem, we conduct a theoretical analysis on the upper bound of the empirical risk in the target domain. Our findings reveal that three key concepts influence the target domain's empirical risk in SFDA scenarios: **1) Representativeness:** ensuring that the selected target data with reliable pseudo-labels sufficiently represents the global distribution of the target domain, **2) Generalization:** measuring the target intra-domain divergence between the selected target data and the remaining unselected data, and

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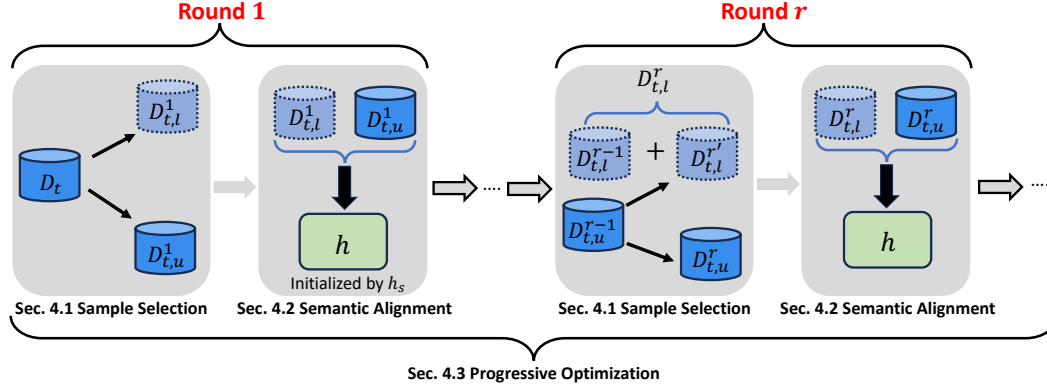


Figure 1. The approach follows a progressive training process (described in Sec. 4.3) conducted over R rounds, with each round comprising Q epochs. Each round consists of two main stages: sample selection (detailed in Sec.4.1) and semantic alignment (discussed in Sec.4.2). The entire unlabeled target dataset is denoted as D_t , with $D_{t,l}^r$ and $D_{t,u}^r$ representing the selected target samples and the remaining unlabeled data during the r -th round, respectively. Specifically, $D_{t,l}^r$ refers to a subset of target samples selected at the r -th round. The model h begins as a pre-trained source domain model h_s and is progressively adapted in each round to enhance target domain performance.

3) Variety: maintaining diversity among the selected data to enhance the model’s adaptability. In this work, we translate these theoretical insights into a practical algorithm to address SFDA more effectively. As shown in Figure 1, first, we introduce an empirical sample selection strategy aimed at selecting target samples that maintain high pseudo-label quality and accurately represent the distributional characteristics of the target domain. Second, we propose a certainty-guided semantic alignment method to minimize intra-domain divergence, ensuring consistency between the selected and remaining target samples. Lastly, we develop a progressive optimization framework that integrates these concepts to facilitate iterative and adaptive training, leading to improved domain adaptation performance.

Our contributions are outlined as follows:

- We conduct a theoretical analysis of the upper bound of target domain empirical risk in the context of SFDA.
- We uncover three pivotal factors—representativeness, generalization, and variety—that influence the control of the upper bound in SFDA scenarios.
- We develop a practical SFDA algorithm that effectively integrates these theoretical insights for enhanced adaptation performance.
- We validate our approach through extensive experiments on standard benchmarks, showcasing significant improvements over previous state-of-the-art methods.

2. Related Works

Unsupervised Domain Adaptation (UDA). Seminal works [1] offer theoretical guarantees for quantifying the discrepancy between the source and target domains. The following works aim to either learn domain-invariant features or leverage adversarial learning to align and reduce the domain discrepancy. For example, Deep Adaptation Network

(DAN) [19] proposes to minimize the multi-kernel maximum mean discrepancy (MK-MMD) to reduce domain drift. Zhang et al. [41] introduce Margin Disparity Discrepancy (MDD) to measure the divergence across domains. Adversarial Discriminative Domain Adaptation (ADDA) [29] combines discriminative modeling, untied weight sharing, and a GAN loss to learn generalizable features across domains. Conditional Domain Adversarial Networks (CDAN) [20] aligns features across domains by exploiting discriminative information conveyed in the classifier predictions. Worth noting that all the methods need both source and target domain data for adaptation.

Source-Free Domain Adaptation (SFDA). An early approach, Source Hypothesis Transfer (SHOT)[15], introduces information maximization and clustering-based pseudo-labeling to encourage confident predictions on the target domain. Semantic Consistency on the Nearest Neighborhood (SCNNH)[27] extends this idea by utilizing estimated cluster centroids to guide semantic consistency. Subsequent efforts in Source-Free Domain Adaptation (SFDA) largely fall into two categories. The first focuses on generating data to mimic the source domain, as seen in Source Data-free Domain Adaptation (SDDA)[12]. Distribution Estimation (DE)[4] further explores this by estimating class-conditioned feature distributions using target data and anchor points. The second direction is self-training based on pseudo-labels provided by the pre-trained model, including methods like Neighborhood Reciprocity Clustering (NRC)[38] and Attracting and Dispersing (AaD)[39]. Recently, [17] improved pseudo-label quality by measuring label uncertainty, and a theoretical analysis of SFDA [21] examined the problem from the perspective of training objectives and existing loss functions. Distinct from these approaches, our theoretical analysis starts from a broader perspective, specifically the general empirical risk of the unlabeled target domain. We

derive an upper bound for this risk, which informs the design of our method. Based on this analysis, we propose a practical algorithm that divides the target data into two subsets: **one subset with high-quality pseudo-labels that preserves the global distribution structure of the target domain, and the remaining data.** Instead of addressing source-target divergence as prior SFDA methods have, we focus on **semantic alignment within the target domain, aligning the two subsets to address intra-domain divergence.**

Data Sampling (DS). Data sampling selects representative data from the entire dataset to improve model performance. For example, AUTO [37] uses an in-out-aware filter to assign pseudo-labels, and ISDA [35] introduces a loss function based on feature norm and prediction entropy to select informative data. In SFDA, several strategies have been proposed. SHOT++ [16] partitions the target dataset into pseudo-labeled and unlabeled subsets based on prediction entropy. ProxyMix [5] selects N nearest target samples to the source class centers as labeled data. BETA [36] divides the target dataset into easy and hard subsets based on adaptation probabilities. Unlike traditional SFDA methods that split the target domain early, often introducing noise due to poor adaptation, we propose a progressive framework. Our method iteratively selects high-confidence pseudo-labeled data, refining the labeled subset over time to improve adaptation and minimize label noise.

3. Investigating SFDA Problem

In this section, we begin by defining the Source-Free Domain Adaptation (SFDA) problem and proceed to present a theoretical analysis of the general empirical risk in the target domain. This analysis highlights key factors that can inform the design of SFDA methods, emphasizing representativeness, generalization, and diversity.

3.1. Problem Formulation

Let $\mathcal{X}$ denote the input space, and $\mathcal{Y} = \{1, \dots, C\}$ denote the label space. A domain is defined by a joint distribution P_{XY} over the instance space $\mathcal{X} \times \mathcal{Y}$. Given a model $h_s : \mathcal{X} \rightarrow \mathcal{Y}$ trained on labeled source domain data $D_s = \{(x_s^i, y_s^i)\}_{i=1}^{n_s}$, h_s represents a function predicting the conditional distribution $P_{Y|X}$. The model h_s can be decomposed into a feature extractor $f_s : \mathcal{X} \rightarrow \mathcal{E}$ and a classifier $g_s : \mathcal{E} \rightarrow \mathcal{Y}$, where $\mathcal{E}$ is the feature embedding space. Thus, the prediction for an input x using h_s can be expressed as $h_s(x) = (g_s \circ f_s)(x)$. For any feature embedding $e \in \mathcal{E}$, the classifier $g_s(e)$ induces a conditional distribution $P_{Y|e}$. The goal of SFDA is to train the model h_s with an unlabeled target domain data $D_t = \{x_t^i\}_{i=1}^{n_t}$ to obtain a target-adapted model h that minimizes the population risk of the target domain, i.e., $\mathbb{E}_{P_{XY}^t}[\ell(h(X), Y)]$ where $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_+$ is a loss function.

3.2. Theoretical Analysis and Insights

Given the above setup, we define the general empirical risk objective for the target domain data $x_t \in D_t$ between the target-adapted model h and the ground truth model $\hat{h}$ as:

$$\ell(h, D_t) = \mathbb{E}_{x_t \in D_t} [|h(x_t) - \hat{h}(x_t)|], \quad (1)$$

where $\hat{h}(x_t)$ represents the ground truth prediction for x_t .

Before presenting the main theorem central to our theoretical analysis, we introduce the following assumptions: for some unlabeled target samples $x_t^i \in D_t$, the model h can produce highly accurate pseudo-labels $\hat{y}_t^i$. We denote this subset of pseudo-labeled data as $D_{t,l}$, while the remaining unlabeled portion is denoted as $D_{t,u}$. Consequently, Eq. 1 can be decomposed as:

$$\ell(h, D_t) \leq \ell(h, D_{t,l}) + \ell(h, D_{t,u}), \quad (2)$$

where $\ell(h, D_{t,l})$ and $\ell(h, D_{t,u})$ denote the empirical risk on $D_{t,l}$ and $D_{t,u}$, respectively.

Inspired by the theoretical framework in [1], we derive the following theorem to provide insight into the SFDA problem (proved in Appendix):

Theorem 1. *Let $\ell(\cdot, \cdot)$ be a loss function applied to a hypothesis h and a dataset D_t for empirical risk minimization. Assume h , parameterized by θ , is trained on D_t and belongs to a hypothesis space $\mathcal{H}$ with VC-dimension d . With a probability of at least $1 - p$ over the sample selection process, the following bound holds:*

$$\begin{aligned} \ell(h, D_t) \leq & \underbrace{2\ell(h, D_{t,l})}_{\text{Representativeness}} + \underbrace{\frac{1}{2} \hat{d}_{\mathcal{H}\Delta\mathcal{H}}(D_{t,l}, D_{t,u})}_{\text{Generalization}} \\ & + 2 \underbrace{\sqrt{\frac{d \log(2m) + \log(\frac{2}{p})}{m}}}_{\text{Variety}} + \lambda \end{aligned} \quad (3)$$

where $\hat{d}_{\mathcal{H}\Delta\mathcal{H}}(\cdot, \cdot)$ denotes the empirical distribution divergence, m indicates the size of $D_{t,l}$, λ indicates the classification error on $D_{t,l}$.

Theorem 1 offers critical insights into controlling the general empirical risk for the unlabeled target domain in SFDA. The first term, $\ell(h, D_{t,l})$, reflects **representativeness**, where $D_{t,l}$ captures the global distribution structure of D_t . These samples typically have high-quality pseudo-labels and are close to the class anchors. The second term, $\hat{d}_{\mathcal{H}\Delta\mathcal{H}}(D_{t,l}, D_{t,u})$, measures the **generalization** capability by quantifying the distribution divergence between $D_{t,l}$ and $D_{t,u}$, which provides new insights to measure the domain divergence for SFDA from target domain instead of source and target domains where source domain is not available. The final term relates to **variety**, influenced by the size of

$D_{t,l}$, which affects the stability of the risk estimation. These terms collectively suggest that the SFDA problem can be addressed through a focus on representativeness, generalization, and variety, ensuring that the learned model maintains an upper bound on the empirical risk in the target domain.

Discussion. In contrast to the existing theoretical analysis of SFDA by Mitsuzumi et al. [21], which focused on the relationship between the training objective and the SFDA training loss, decomposing the loss into components of discriminability and variety, our analysis approaches the problem from a broader perspective. Specifically, we begin with the general empirical risk of the unlabeled target domain, independent of any pre-existing SFDA training objectives. This allows for a more comprehensive understanding of how representativeness, generalization, and variety contribute to controlling the empirical risk in SFDA scenarios.

4. Theoretically Inspired Practical Algorithm

The theoretical analysis above provides a foundation for designing strategies to leverage representativeness, generalization, and variety to optimize the general empirical risk in the target domain, as outlined in Theorem 1. Building on these insights, we introduce a practical SFDA algorithm that incorporates these concepts. First, we devise a surrogate selection strategy for pseudo-labeled target samples $D_{t,l}$, to enhance **representativeness**, ensuring that these samples accurately reflect the global distribution of D_t . Next, we implement a semantic alignment mechanism to address the **generalization** term, minimizing target intra-domain divergence between $D_{t,l}$ and the unlabeled portion $D_{t,u}$. Lastly, we propose a progressive training framework to handle **variety**, improving the stability and robustness of risk estimation within $D_{t,l}$. This multi-faceted approach aligns with the theoretical underpinnings and promotes a more effective solution to the SFDA problem.

4.1. Intersection based Sample Selection

A straightforward sample selection mechanism involves utilizing the classifier f within the model h to assign class labels to target samples and subsequently perform top- k sample selection, as detailed below.

$$D_{t,l}^C = \bigcup_{c=1}^C \{\text{Top}_k(D_t^c)\}, \text{ with } \text{Top}_k(D_t^c) = \{x_t^i | i \in \text{argsort}(h(x_t^i)), x_t^i \in D_t^c, i \leq k\}. \quad (4)$$

In this context, $D_{t,l}^C$ denotes the selected samples, D_t^c represents the subset of unlabeled target data assigned a pseudo-label $\tilde{y}_{t,l}^i = c$, and k denotes the number of selected samples. We refer to this method as classifier-based sampling (**C-sampling**). However, the standard C-sampling approach has two significant limitations: (1) it inherits labeling biases from the source-domain pre-trained model h_s , which can

lead to inaccurate pseudo-labels, and (2) the high-confidence samples selected are often confined to the distribution of the source domain. This occurs because deep learning classifiers are inherently data-driven and tend to memorize training distributions, thereby exacerbating the distribution gap between $D_{t,l}$ and $D_{t,u}$, increasing domain drift.

To mitigate the above issues, we leverage the unsupervised information by clustering the centers populated from the unlabeled target domain samples, which faithfully reflects the target domain distribution and pops up the most confident samples for sample selection. We term it as target class center based sampling (**T-sampling**). Specifically, we apply the softmax over the logits after classifier f to gauge the likelihood of correct classification, and select the top $\sigma\%$ of target data per class denoted as $D_{t,l}^{c,\sigma}$, to form a reliable set. This set is then utilized to estimate the target class centers through a weighted aggregation process:

$$\mu_c = \frac{\sum_{x_t^i \in D_{t,l}^{c,\sigma}} p_t^{i,c} \cdot z_t^i}{\sum_{x_t^i \in D_{t,l}^{c,\sigma}} p_t^{i,c}}, \quad (5)$$

where $p_t^{i,c}$ is the probability of the x_t^i belongs to c -th class and z_t^i is the feature of x_t^i . Following this, we calculate the cosine similarity between the estimated class centers μ_c and the unlabeled target data, and proceed to select data with high cosine similarity from each class:

$$D_{t,l}^T = \bigcup_{c=1}^C \{\text{Top}_k^{\cos}(D_{t,u})\}, \text{ with } \text{Top}_k^{\cos}(D_{t,u}) = \{x_t^i | i \in \text{argsort}(\cos(x_t^i, \mu_c)), x_t^i \in D_{t,u}, i \leq k\}, \quad (6)$$

where $D_{t,l}^T$ denotes the selected samples.

Intuitively, the closer the unlabeled samples are to the estimated class centers, the higher the likelihood these samples belong to the specific classes. However, it is worth noting that the accuracy of the estimated class centers heavily relies on the pseudo label information generated by the model updated from the initial training rounds. This accuracy can be compromised due to inherent domain drift issues.

To further improve the pseudo-labeling accuracy, we empirically introduce an intersection based sampling, denoted as **I-sampling**, to retrieve the intersection between C-sampling and T-sampling.

$$D_{t,l}^I = \bigcup_{c=1}^C \{\text{Top}_k(D_{t,l}^c) \cap \text{Top}_k^{\cos}(D_{t,u})\}. \quad (7)$$

In Section 5.3, we empirically validate that **I-sampling** selects unlabeled target data with consistently higher pseudo-label purity compared to C-sampling and T-sampling.

The supervised cross-entropy loss is then applied to I-sampling selected data to update the model h :

$$\mathcal{L}_{CE}(h; D_{t,l}) = -\mathbb{E}_{(x_{t,l}^i, \tilde{y}_{t,l}^i) \in D_{t,l}} \sum_{c=1}^C \tilde{y}_{t,l}^{i,c} \log \sigma_c(h(x_{t,l}^i)), \quad (8)$$

where σ is the softmax operation, σ_c indicates the c -th element in the softmax output of the model, and $\tilde{y}_{t,l}^{i,c}$ denotes the c -th element in one-of- C encoding of $\tilde{y}_{t,l}^i$.

4.2. Semantic Target Intra-Domain Alignment

In this section, we focus on minimizing the intra-domain divergence between $D_{t,l}$ and $D_{t,u}$ within the target domain. While adversarial feature alignment techniques have proven effective in various scenarios [7], directly applying them in SFDA presents challenges due to the typical size imbalance between $D_{t,l}$ and $D_{t,u}$. This imbalance can introduce negative effects, complicating domain alignment efforts. To address this, we propose a straightforward yet effective semantic alignment approach. Our method incorporates a pseudo-label refining technique to enhance the labeling quality for $D_{t,u}$ and employs a certainty-weighted strategy to facilitate more reliable semantic alignment between $D_{t,l}$ and $D_{t,u}$.

Pseudo-Label Refining: To refine the labels for an unlabeled target sample $x_{t,u}^i$ drawn from $D_{t,u}$, we establish two memory banks: one for storing target features and another for storing logits. These memory banks are updated dynamically throughout the training process, with new features and their corresponding logits added from each mini-batch. For any given $x_{t,u}^i$, its feature representation $z_{t,u}^i$ is used to find the $\mathcal{M}$ nearest neighbors from target feature memory bank. The label for $x_{t,u}^i$ is subsequently refined by aggregating the logits from the selected $\mathcal{M}$ nearest neighbors, employing a cosine similarity-weighted mechanism to enhance the accuracy and reliability of the pseudo-label:

$$\tilde{p}_{t,u}^i = \sum_{m=1}^{\mathcal{M}} \alpha^m p_t^m \text{ and } \alpha^m = \frac{\exp(z_{t,u}^i \cdot z_t^m / \tau)}{\sum_{j=1}^{\mathcal{M}} \exp(z_{t,u}^i \cdot z_t^j / \tau)}, \quad (9)$$

where z_t^m denotes the feature of the nearest neighbors alone with its logits p_t^m . α^m measures the cosine similarity between the current data $x_{t,u}^i$ and the m -th nearest neighbor x_t^m .

Certainty Weighted Semantic Alignment: The iterative nature of the refinement process means that the pseudo-labels produced through neighborhood label propagation may still contain noise. To address this, we introduce a harmonic certainty metric inspired by [6], defined as:

$$\eta(\tilde{p}_{t,u}^i) = \frac{(1 - H(\tilde{p}_{t,u}^i)) + \text{Max}(\log(\tilde{p}_{t,u}^i))}{2}, \quad (10)$$

where $H(\cdot)$ denotes the entropy calculation and $\text{Max}(\cdot)$ indicates the maximum probability value in $\log(\tilde{p}_{t,u}^i)$. We normalize the $H(\tilde{p}_{t,u}^i)$ and $\text{Max}(\log(\tilde{p}_{t,u}^i))$ by minmax normalization to unify them within $[0, 1]$ before computing $\eta(\tilde{p}_{t,u}^i)$. This metric is designed to trade off the two ends. One is estimating certainty solely based on entropy may falter in

distinguishing between confident and extremely sharp predictions. The other is relying solely on probability for certainty estimation may not effectively address the challenge of distinguishing between various class distributions. With the refined label information $\tilde{p}_{t,u}^i$ and its certainty estimation $\eta(\tilde{p}_{t,u}^i)$, we align the domain drift between $D_{t,l}$ and $D_{t,u}$ by a semantic alignment regularization, which pushes the data augmented $\text{Aug}(x_{t,u}^i)$ to the corresponding class centers learned by the model h :

$$\mathcal{L}_{SA}(h; D_{t,u}) = \mathbb{E}_{x_{t,u}^i \in D_{t,u}} \mathbb{I}(\eta(\tilde{p}_{t,u}^i) \geq \beta) \phi(h(\text{Aug}(x_{t,u}^i)), w^{\tilde{y}_{t,u}^i}), \quad (11)$$

where β is the threshold, $\phi(\cdot, \cdot)$ is any differentiable distance measure. w is the row-wise neuron weights from the classifier f viewed as the model learned class centers, and $\tilde{y}_{t,u}^i = \arg \max(\tilde{p}_{t,u}^i)$ is the label information for $x_{t,u}^i$.

During training, the proposed regularization technique facilitates a self-paced curriculum learning process. Initially, the model may produce low-quality label predictions for $x_{t,u}^i$. By employing the threshold parameter β , samples with uncertain or noisy label information are effectively filtered out, reducing the risk of negative transfer and ensuring that only reliable data contribute to training. As the training advances, the model progressively refines its representation, enabling more consistent alignment of the unlabeled target samples with their corresponding class centers. This leads to the generation of higher-quality label predictions over time, enhancing the semantic alignment between $D_{t,l}$ and the remaining unlabeled target data $D_{t,u}$, and ultimately promoting better domain adaptation.

Algorithm 1 Theoretical Insights Guided Method for SFDA

Input: Unlabeled target domain dataset D_t , initialized model h by source domain pretrained model h_s .
Output: Learned model h .
With E Epochs
for $e = 1$ to E do
 # Q epochs interval
 if $\text{Mod}(e, Q) = 0$ then
 Apply Equation 7 for target data selection (Sec. 4.1).
 end if
 # I is the length of D_t
 for i to I do
 Apply overall loss Equation 13 to train h (Sec. 4.3).
 end for
end for

4.3. Progressive Optimization Regime

To prevent posterior collapse and ensure meaningful feature learning, we follow the standard state-of-the-art protocol by optimizing the following Information Maximization (IM)

regularisation:

$$\mathcal{L}_{IM}(h; D_{t,u}) = \sum_c P_c \log(P_c) - \mathbb{E}_{x_{t,u}^i \in D_{t,u}} \sum_{c=1}^C \sigma_c(h(x_{t,u}^i)) \log(\sigma_c(h(x_{t,u}^i))), \quad (12)$$

where $P_c = \mathbb{E}_{x_{t,u}^i \in D_{t,u}} [\sigma_c(h(x_{t,u}^i))]$. The overall objective for source-free domain adaptation is:

$$\mathcal{L}_{SFDA} = \mathcal{L}_{CE} + \mathcal{L}_{SA} + \rho \mathcal{L}_{IM}, \quad (13)$$

where ρ is the loss balancing hyper-parameter.

Throughout the training process, a natural curriculum is established, guided by the $\mathcal{L}_{SFDA}$ loss. As training progresses, the model h is able to generate higher-quality label information for $D_{t,u}$ compared to the pre-trained model h_s . Building on this observation, we propose a progressive training framework, where we dynamically select samples from $D_{t,u}$ to augment $D_{t,l}$ using I-sampling, after every Q epochs. This framework effectively balances the variety within $D_{t,l}$ and the risk induced by noisy pseudo-labels. Algorithm 1 illustrates details of our proposed method.

5. Experiment

5.1. Experimental Settings

Datasets: We demonstrate our method under image classification task on three standard domain adaptation benchmarks as most of the literature does: Office-Home [30], DomainNet126 [23], and VisDA-C 2017 (VisDA-C) [22]. **(1) Office-Home** consists of four domains (Art, Clipart, Produce, and Real-world) with 65 classes. **(2) DomainNet126** is one of the largest domain adaptation datasets which contains 6 domains and 345 classes. Following the protocol in [17], we use a subset including 126 classes from 4 domains (Clipart, Painting, Real, Sketch), and evaluate on 7 designed domain tasks. **(3) VisDA-C** is a challenging large-scale synthesis-to-real object recognition dataset that contains 12 classes. The source domain includes 152k synthetic images and the target domain contains 55k real images.

Baselines: We compare to three main streams of methods that are most relevant to ours: (1) *Unsupervised Domain Adaptation* methods, including CDAN [20], MCC [9], and CST [18]. (2) *Source-Free Domain Adaptation* methods, including SHOT [15], A²Net [33], NRC [38], AaD [39], U-SFAN [25], CoWA [13], SDE [4], DaC [42], GPUE [17], C-SFDA [10], I-SFDA [21], SHOT+DPC [34], and TPDS [28]. (3) *Test Time Adaptation* methods, including TENT [31] and AdaCon [2]. Besides, we include Empirical Risk Minimization (ERM) [11] as a general baseline.

Evaluation Metrics: For Office-Home and DomainNet126, we report the top-1 accuracy under each domain task together with their average. For VisDA-C, we report the per-class top-1 accuracy and their average. Each domain task is conducted independently 3 times and average is reported.

Implementation Details: For a fair comparison, we adopt the pretraining methodology outlined in SHOT [15] to get the source domain pre-trained model. In particular, we separately employ ResNet50 for Office-Home and DomainNet126, and ResNet101 for VisDA-C as the backbone architectures. Subsequently, we replace the original final FC layer with a new bottleneck layer, followed by Batch Normalization (BN). In the target domain adaptation phase, we initialize the target model using the parameters from the pretrained source domain model. We utilize SGD with momentum 0.9 and weight decay $1e^{-3}$ and batch size of 64 for all benchmarks. The initial learning rates are 0.001 for Office-Home and DomainNet126 and 0.01 for VisDA-C, respectively. We set $\rho = \exp(\text{ite}/\text{max_ite})^{-1}$ where max_ite is the maximum number of the training iterations. More implementation details can be found in Appendix.

5.2. Experimental Results

Table 1 illustrates the quantitative comparison on Office-Home and VisDA-C datasets. The methods are organized into the top three rows, while ours is in the last row. The first row is the ERM result, which serves as the lower bound. The second row shows three representative unsupervised domain adaptation (UDA) methods, all of which necessitate the availability of the source domain dataset during adaptation to the target domain. The third row contains seven cutting-edge source-free domain adaptation (SFDA) methods, such as the I-SFDA and SHOT+DPC. For Office-Home, under the UDA setting, our approach surpasses the strong baseline CST by +1.1% and outperforms all other UDA baselines by an even larger margin. This achievement is noteworthy considering that we do not utilize any source data during source-free adaptation. In the more challenging SFDA setting, we achieve a 2.2% improvement over U-SFAN, a 1.2% improvement over SDE, and a 0.9% improvement over SHOT+DPC in terms of the ‘Avg.’ metric. Furthermore, our method outperforms all other SFDA baselines in 8 out of 12 domain tasks, firmly establishing its superiority. Similar trends are observed on VisDA-C, further validating the effectiveness of our approach.

The evaluation results on DomainNet126 are detailed in Table 2. We adhere to the experimental protocol outlined in [17], which allows us to showcase the performance of our method across 7 different domain tasks. By comparing our method with both UDA and SFDA baselines, our method achieves state-of-the-art performance among all baselines and is higher than the second best GPUE by a margin of 3.6% in terms of ‘Avg.’. Furthermore, our method attains the highest accuracy across all 7 domain tasks. The achievement not only demonstrates the superiority of our proposed method but also underscores the effectiveness of our proposed approach in addressing source-free domain adaptation within the context of a large-scale benchmark.

Table 1. Classification Accuracy (%) on Office-Home (ResNet-50) and VisDA-C (ResNet-101). The best results under SFDA setting are highlighted in bold. Note that “SF” means whether a method belongs to SFDA method.

Method	SF	Office-Home													VisDA-C Avg.
		A→C	A→P	A→R	C→A	C→P	C→R	P→A	P→Cl	Pr→R	R→A	R→C	R→P	Avg.	
ERM	×	34.9	50.0	58.0	37.4	41.9	46.2	38.5	31.2	60.4	53.9	41.2	59.9	46.1	52.4
CDAN [20]	×	50.7	70.6	76.0	57.6	70.0	70.0	57.4	50.9	77.3	70.9	56.7	81.6	65.8	73.9
CST [18]	×	59.0	79.6	83.4	68.4	77.1	76.7	68.9	56.4	83.0	75.3	62.2	85.1	73.0	-
SHOT [15]	✓	57.1	78.1	81.5	68.0	78.2	78.1	67.4	54.9	82.2	73.3	58.8	84.3	71.8	82.9
A ² Net [33]	✓	58.4	79.0	82.4	67.5	79.3	78.9	68.0	56.2	82.9	74.1	60.5	85.0	72.8	84.3
AaD [39]	✓	59.3	79.3	82.1	68.9	79.8	79.5	67.2	57.4	83.1	72.1	58.5	85.4	72.7	88.0
CoWA [13]	✓	56.9	78.4	81.0	69.1	80.0	79.9	67.7	57.2	82.4	72.8	60.5	84.5	72.5	86.9
DaC [42]	✓	59.1	79.5	81.2	69.3	78.9	79.2	67.4	56.4	82.4	74.0	61.4	84.4	72.8	87.3
C-SFDA [10]	✓	60.3	80.2	82.9	69.3	80.1	78.8	67.3	58.1	83.4	73.6	61.3	86.3	73.5	87.8
I-SFDA [21]	✓	60.7	78.9	82.0	69.9	79.5	79.7	67.1	58.8	82.3	74.2	61.3	86.4	73.4	88.4
SHOT+DPC [34]	✓	59.2	79.8	82.6	68.9	79.7	79.5	68.6	56.5	82.9	73.9	61.2	85.4	73.2	84.3
Ours	✓	61.2	80.9	82.7	69.3	81.2	81.4	68.1	58.8	83.4	74.6	62.4	85.7	74.1	89.1

Table 2. Classification Accuracy (%) on DomainNet-126 (ResNet-50). The best results under SFDA setting are highlighted in bold. Note that “SF” means whether a method belongs to SFDA method.

Method	SF	Re→Cl	Re→Pa	Pa→Cl	Cl→Sk	Sk→Pa	Re→Sk	Pa→Re	Avg.
ERM	×	55.5	62.7	53.0	46.9	50.1	46.3	75.0	55.6
MCC [9]	×	44.8	65.7	41.9	34.9	47.3	35.3	72.4	48.9
SHOT [15]	✓	67.7	68.4	66.9	60.1	66.1	59.9	80.8	67.1
TENT [31]	✓	58.5	65.7	57.9	48.5	52.4	54.0	67.0	57.7
NRC [38]	✓	64.7	69.4	62.9	58.3	65.8	58.7	81.3	65.9
CoWA [13]	✓	69.0	67.2	66.2	60.6	65.8	60.0	79.8	66.9
AdaCon [2]	✓	70.2	69.8	68.6	58.0	65.9	61.5	80.5	67.8
GPUE [17]	✓	74.2	70.4	68.8	64.0	67.5	65.7	76.5	69.6
TPDS [28]	✓	66.4	67.0	65.6	59.8	64.3	61.5	79.0	66.2
Ours	✓	77.6	72.9	71.9	68.0	71.2	67.3	83.2	73.2

5.3. Model Analysis

Ablation Study on Equation (13). Table 3 shows the ablative results regarding the proposed modules. It is organized into three rows. In the first row, we individually explore each module within our method. Notably, the $\mathcal{L}_{SA}$ module exhibits the lowest accuracy at 28.2% compared to the other two modules. This discrepancy arises from the fact that training the model exclusively with the selected $D_{t,l}$ dataset or failing to prevent poster collapse would lead to degradation from the original source domain pre-trained model. This is because $D_{t,l}$ is relatively small and contains incorrect label information. In the second row, we consistently observe improvements in accuracy when randomly combining two sub-modules. Notably, when we combine $\mathcal{L}_{SA}$ with either $\mathcal{L}_{CE}$ or $\mathcal{L}_{IM}$, we achieve a significant accuracy boost by more than 40.0%. Finally, in the last row, we showcase the performance enhancement achieved by our full approach, which further improves accuracy by more than 2.0%.

Analysis of Accuracy in Figure 1. We explore the accuracy of the target data from different subsets at r -th round, i.e., all target data D_t , selected target data $D_{t,l}^{r'}$, all selected target data $D_{t,l}^r$, and remaining target data $D_{t,u}^r$. Figure 2 (a) presents the overall trend on D_t . To delve into the source of this advantage, we further examine the accuracy trend of two aspects: the selected target data $D_{t,l}^{r'}$ at each round, and the overall selected target data $D_{t,l}^k$. As depicted in Figures 2 (b) and (c), we observe that **I-sampling** strategy excels in selecting target data with more accurate labels, thereby enhancing model training. It’s worth noting that as the data sampling process continues, the accuracy of selected target data with labels tends to decrease. This arises from the fact that the most reliable target data with pseudo-labels are chosen in the initial stages. Additionally, we see that the decline in accuracy for the selected target data $D_{t,l}^{r'}$ during the initial phase is less pronounced compared to the later stages. Conversely, the increase in accuracy for the entire target data D_t

Table 3. Ablation study of sub-modules in our proposed method on Office-Home and DomainNet126 datasets with six domain tasks, i.e., $A \rightarrow C$, $A \rightarrow P$, $A \rightarrow R$, $Re \rightarrow Cl$, $Re \rightarrow Pa$, and $Re \rightarrow Sk$. $\mathcal{L}_{CE}$ is the cross-entropy loss for $D_{t,l}$. $\mathcal{L}_{SA}$ is the semantic domain alignment loss for $D_{t,u}$. $\mathcal{L}_{IM}$ is the information maximization loss for $D_{t,u}$.

$\mathcal{L}_{SA}$	$\mathcal{L}_{CE}$	$\mathcal{L}_{IM}$	$A \rightarrow C$	$A \rightarrow P$	$A \rightarrow R$	$Re \rightarrow Cl$	$Re \rightarrow Pa$	$Re \rightarrow Sk$	Avg.
×	×	✓	55.9	75.4	77.6	66.8	68.1	59.0	67.1
×	✓	×	56.9	73.7	78.5	66.1	62.5	57.2	65.8
✓	×	×	12.7	33.3	40.7	24.2	38.6	19.9	28.2
×	✓	✓	58.7	75.7	81.1	70.2	67.9	60.7	69.0
✓	×	✓	56.3	76.7	80.6	69.5	66.6	63.5	68.9
✓	✓	×	59.5	78.7	80.5	74.9	69.7	64.4	71.3
✓	✓	✓	61.2	80.9	82.7	77.6	72.9	67.3	73.8

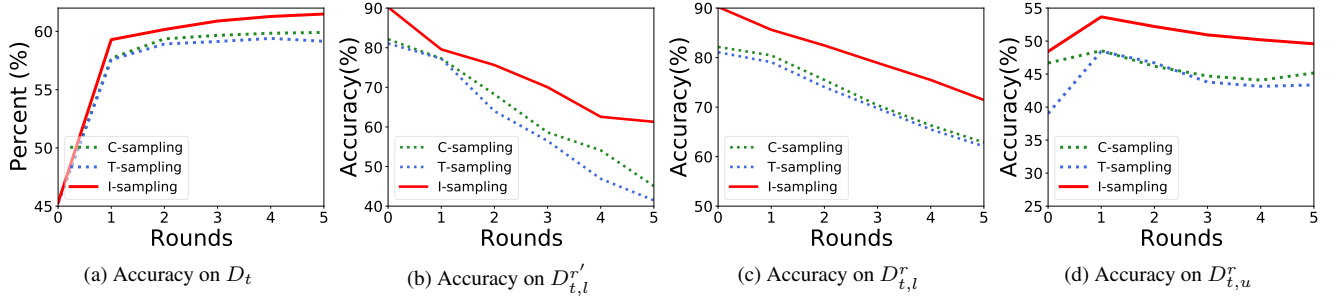


Figure 2. As shown from left to right, four figures illustrate the accuracy of different target subsets during the training process with domain task $A \rightarrow C$ on Office-Home. (a) demonstrates the accuracy of all target data D_t . (b) shows the accuracy of the selected target data $D_{t,l}^{r'}$. (c) describes the accuracy of the all selected target data $D_{t,l}^r$. (d) plots the trend of the accuracy of the remaining target data $D_{t,u}$.

Table 4. Ablation study of different data sampling strategies on two adaptation tasks, i.e., $A \rightarrow C$ (OfficeHome) and $Re \rightarrow Cl$ (DomainNet126).

Sampling strategies	$A \rightarrow C$	$Re \rightarrow Cl$
C-sampling	59.9	76.5
T-sampling	59.1	76.2
I-sampling	61.2	77.7

during the early stages is more substantial than in the later stages. It is likely that accuracy on non-selected target data experiences significant improvement in the early training stages, leading to the selection of more reliable target data for labeling, which is verified in Figure 2 (d).

Ablation Study on Sample Selection in Sec. 4.1. We conduct the study on adaptation tasks $A \rightarrow C$ and $Re \rightarrow Cl$ from OfficeHome and DomainNet126, respectively, to assess the performance of C-sampling, T-sampling, and I-sampling. As shown in Table 4, the I-sampling strategy consistently outperforms both C-sampling and T-sampling across various adaptation tasks, underscoring its advantage compared to the other two sampling strategies.

6. Conclusion

We present a theoretical analysis of the empirical risk bound for the target domain in the Source-Free Domain Adapta-

tion (SFDA) problem. We derive the upper bound for the risk and identify three key factors—representativeness, generalization, and variety—that influence the control of this upper bound. Building on these insights, we propose a practical SFDA algorithm that effectively integrates these factors. Specifically, we introduce an intersection-based sampling strategy to enhance the pseudo-labeling process, focusing on selecting target data with high-quality pseudo-labels. Additionally, we propose a certainty-weighted semantic domain alignment approach to reduce the intra-domain divergence between selected target samples and the remaining target samples. A progressive optimization scheme is employed to iteratively refine these two steps across multiple rounds. Extensive experiments on three major domain adaptation benchmarks demonstrate that our method consistently outperforms both unsupervised domain adaptation and existing source-free domain adaptation methods.

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